LAND USE CONFLICTS

- **AI/ML powered project approaches to resolve land use conflicts**
  - COURSE:BTECH CSE-SECTION-1CSE26
  - SUBJECT:INTRODUCTION TP AI/ML
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EVALUATION-3

➢ STEPS AND ALGORITHM WRITING SKILLS

• STEP-BY-STEP PLAN:

STEP-1: UNDERSTANDING THE PROBLEM

- Gain knowledge of different land use types(Agriculture,urban,forest).
- Understand how and why conflicts arise.EXAMPLE:When factory setup in forest region.

STEP-2:DATA COLLECTION

- Collect data such as:
  - Satellite images and land maps
  - Land type(Urban,agriculture,forest)
  - Areas rich in population,soil fertility,vegetation.
  - Use open datasets from:
    1. Kaggle(land use classification dataset)
    2. Google Earth Engine
    3. ISRO Geoportal(bhuvan.nrsc.gov.in)

STEP-3:DATA PREPROCESSING

- Clean the data (remove wrong or irrelevant data)
- Convert all image to same size.
- Label the image(forest=0,urban=1,agriculture=2)

STEP-4: APPLY MACHINE LEARNING ALGORITHM

- Use Decision tree algorithm for classification:
  - If land is fertile=for agriculture use.
  - If land is in city=for urbanisation.
  - If land is high in vegetation=for forest.
- Use Regression to predict future land change trends.

STEP-5: TRAIN ML MODEL

- Train the decision tree on the training dataset so it can learns the relationship between land features and land type.

STEP-6: VALIDATE ACCURACY

- Run the model on the testing data.
- Validate accuracy(if result is close to actual result).

STEP-7: PREDICT LAND USE CONFLICTS

- If two or more categories overlap(like forest+urban),mark as CONFLICT ZONE.

STEP-8: VISUALISATION

- Display result on maps using Geopandas or folium.
- Highlight conflict areas.

STEP-9:COMPILE RESULT

- Show result in graph or maps (ex-conflicts area in red).
- Show model accuracy

Algorithm

1. **Start**
2. **Input the dataset containing features of land areas (e.g., vegetation index, soil type, , latitude, longitude, etc.) and their land use labels.**
3. **Preprocess the data:**
    - **Handle missing values**
    - **Normalize numerical data**
    - **Split dataset into training (80%) and testing (20%) sets**
4. **Build the Decision Tree Model:**
    - **Select a target variable (e.g., Land_Use_Type).**
    - **For each feature, calculate Information Gain .**
    - **Split the dataset based on data.**
    - **Repeat the process for each category until all data points are classified.**
5. **Train the model using the training dataset**
6. **Test the model using the testing dataset and check accuracy.**
7. **Predict land use type for new/unlabeled areas.**
8. **Compare predicted land use with actual land records**
    - **If Predicted Land Use ≠ Actual Land Use, mark that location as a Conflict Zone**
9. **Visualize Results:**
    - **Display the classified areas and conflict zones on a map or chart**
10. **End**